

measuring the brightness of the lighting of the scene you photograph and adjusting itself to give you the proper exposure. In a digital camera, the image sensor electronically measures brightness at every pixel in the image and arrives at a single exposure value for the entire picture. Let's stop for a bit and understand what exposure values are all about.

### Exposure Value (EV)

Exposure values are just numbers assigned to particular fixed combinations of lens aperture and shutter speed. As seen earlier, there's a range of combinations of shutter speeds and aperture values that yield the same final exposure. Exposure values reduce the need to remember shutter speeds and aperture values and f-numbers. For example, an exposure value zero (EV 0) means an exposure time of 1 second and the aperture value  $f/1.0$ . Similarly, EV 1 is either  $1/2$  seconds and  $f/1.0$  or 1 second and  $f/1.4$ . The table below lists a few familiar EV numbers.

|               |      | Aperture Value |     |     |     |     |     |     |      |    |    |    |    |    |                |  |  |  |  |
|---------------|------|----------------|-----|-----|-----|-----|-----|-----|------|----|----|----|----|----|----------------|--|--|--|--|
| Shutter Speed |      | 1.0            | 1.4 | 2.0 | 2.8 | 4.0 | 5.6 | 8.0 | 11.0 | 16 | 22 | 32 | 45 | 64 | Exposure Value |  |  |  |  |
|               | 1    | 0              | 1   | 2   | 3   | 4   | 5   | 6   | 7    | 8  | 9  | 10 | 11 | 12 |                |  |  |  |  |
|               | 2    | 1              | 2   | 3   | 4   | 5   | 6   | 7   | 8    | 9  | 10 | 11 | 12 | 13 |                |  |  |  |  |
|               | 4    | 2              | 3   | 4   | 5   | 6   | 7   | 8   | 9    | 10 | 11 | 12 | 13 | 14 |                |  |  |  |  |
|               | 8    | 3              | 4   | 5   | 6   | 7   | 8   | 9   | 10   | 11 | 12 | 13 | 14 | 15 |                |  |  |  |  |
|               | 15   | 4              | 5   | 6   | 7   | 8   | 9   | 10  | 11   | 12 | 13 | 14 | 15 | 16 |                |  |  |  |  |
|               | 31   | 5              | 6   | 7   | 8   | 9   | 10  | 11  | 12   | 13 | 14 | 15 | 16 | 17 |                |  |  |  |  |
| Standard      | 60   | 6              | 7   | 8   | 9   | 10  | 11  | 12  | 13   | 14 | 15 | 16 | 17 | 18 |                |  |  |  |  |
|               | 125  | 7              | 8   | 9   | 10  | 11  | 12  | 13  | 14   | 15 | 16 | 17 | 18 | 19 |                |  |  |  |  |
|               | 250  | 8              | 9   | 10  | 11  | 12  | 13  | 14  | 15   | 16 | 17 | 18 | 19 | 20 |                |  |  |  |  |
|               | 500  | 9              | 10  | 11  | 12  | 13  | 14  | 15  | 16   | 17 | 18 | 19 | 20 | 21 |                |  |  |  |  |
|               | 1000 | 10             | 11  | 12  | 13  | 14  | 15  | 16  | 17   | 18 | 19 | 20 | 21 | 22 |                |  |  |  |  |

Metering only creates information. The digital camera allows you to either adjust the settings as you think fit, or does it automatically. However, there are occasions when the camera fails to accurately detect the intensity of light (subject luminance) in all its variations. This may lead to overexposure (the picture appears too light and white) or underexposure (dark all over). This is where exposure compensation helps you out.